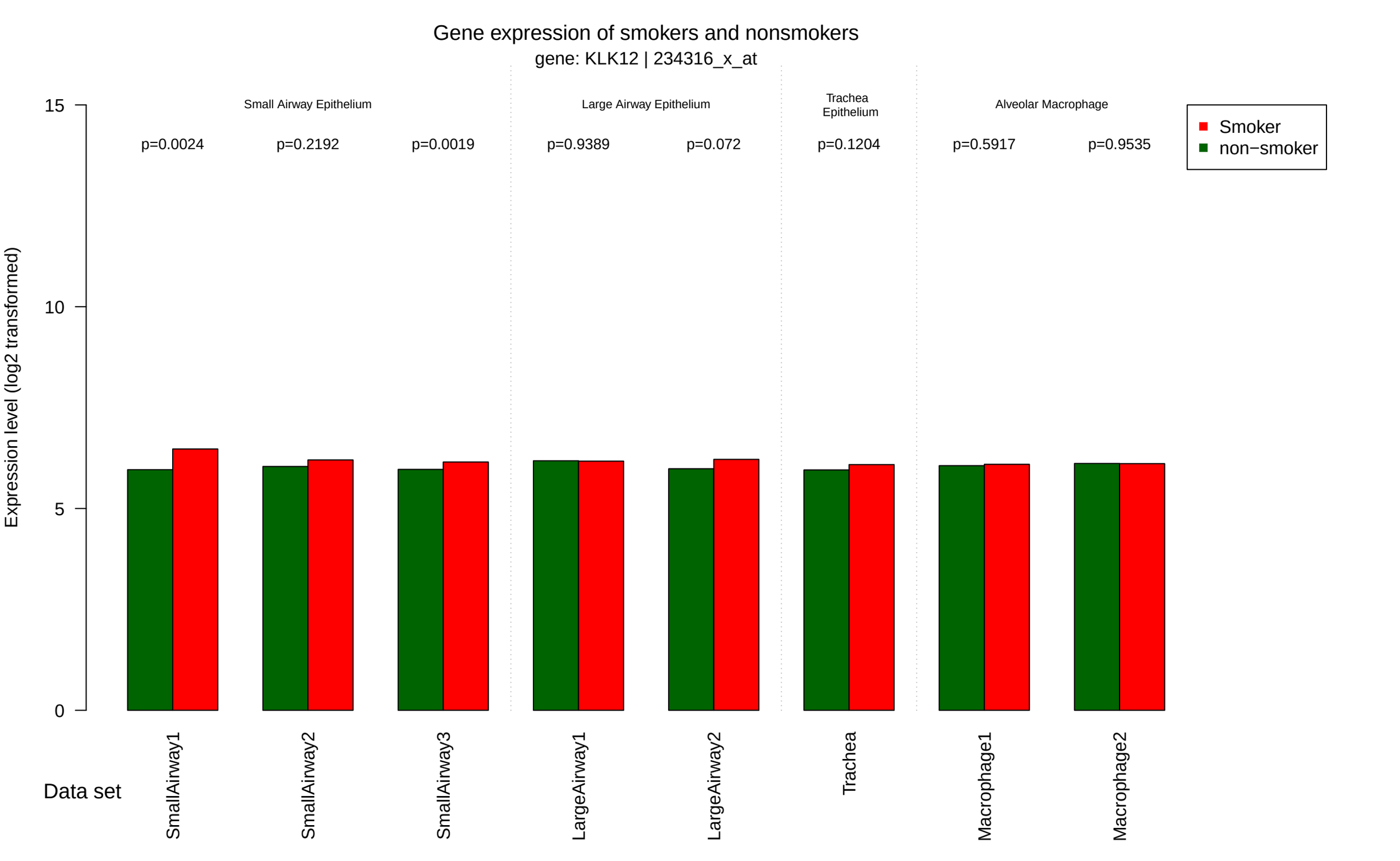
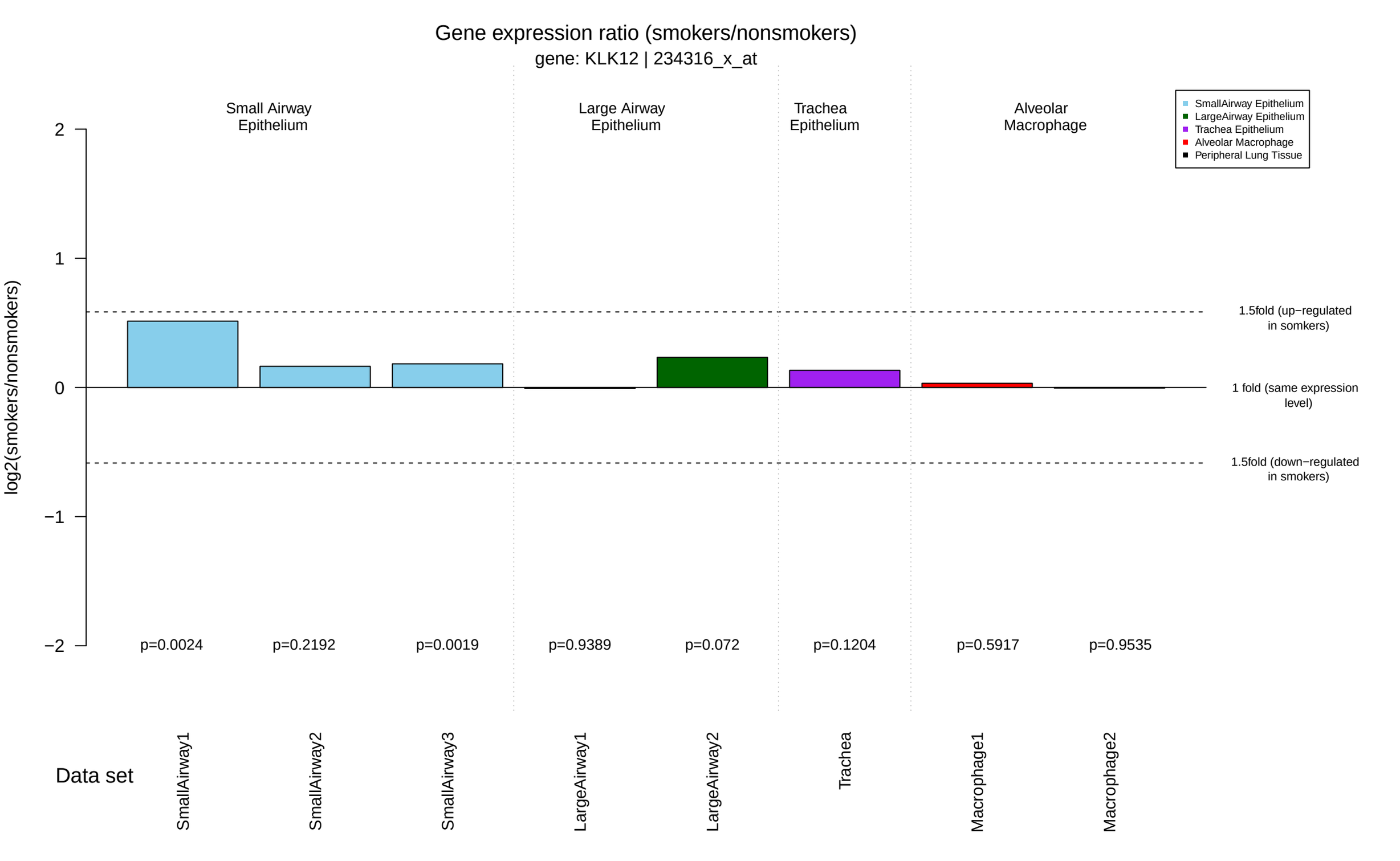
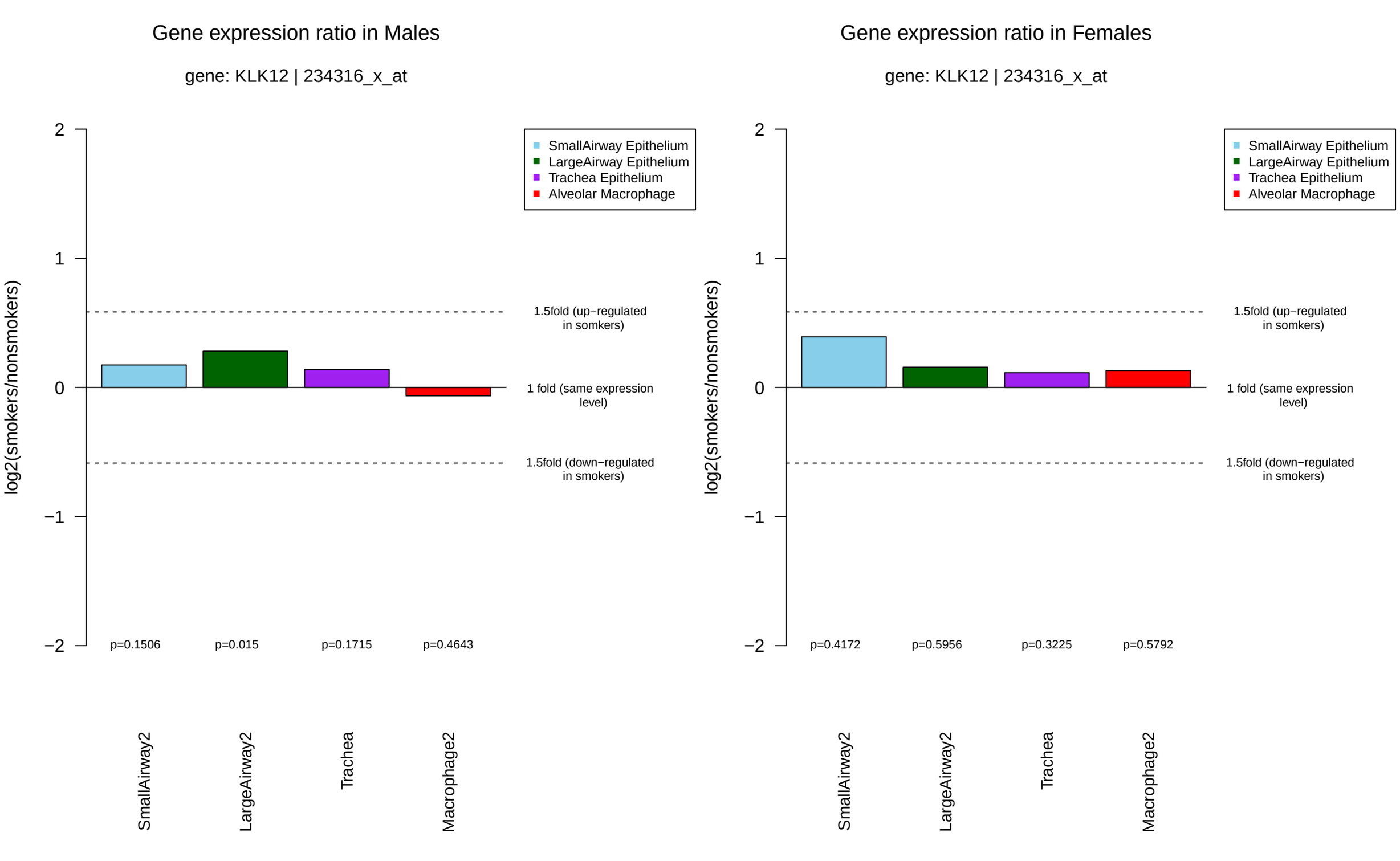
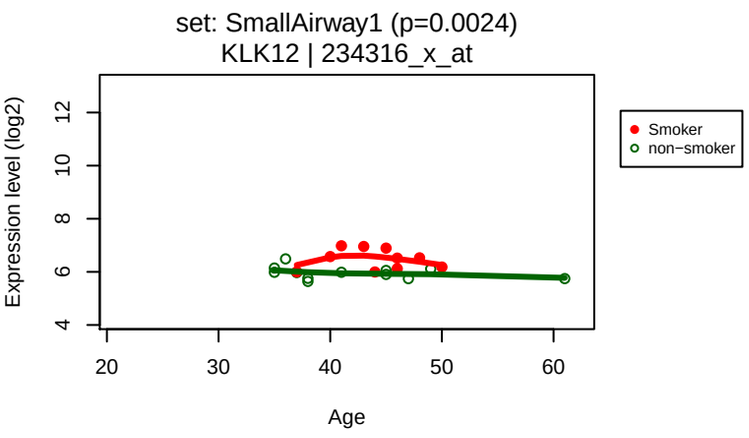
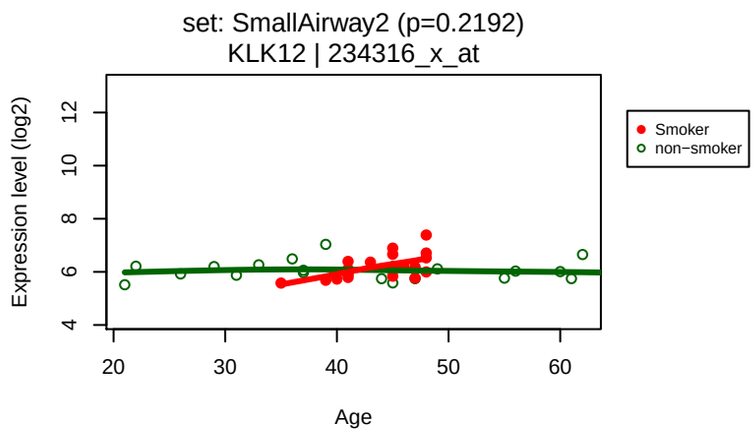
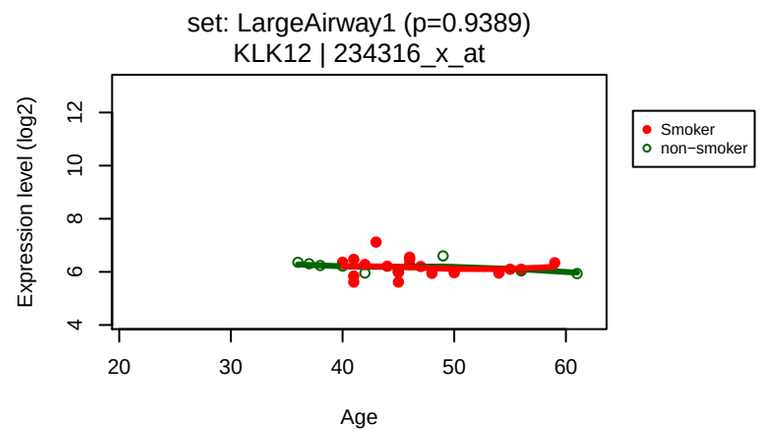

Gene expression of smokers and nonsmokers
gene: KLK12 | 234316_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0024
p=0.2192
p=0.0019
p=0.9389
p=0.072
p=0.1204
p=0.5917
p=0.9535
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

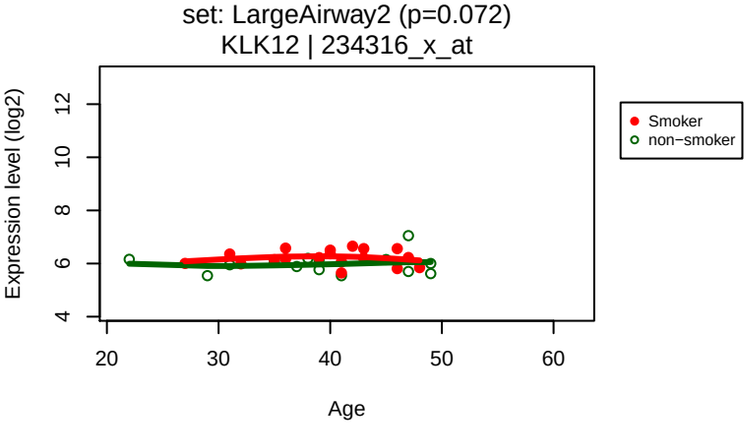

Gene expression ratio (smokers/nonsmokers)
gene: KLK12 | 234316_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0024
p=0.2192
p=0.0019
p=0.9389
p=0.072
p=0.1204
p=0.5917
p=0.9535
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

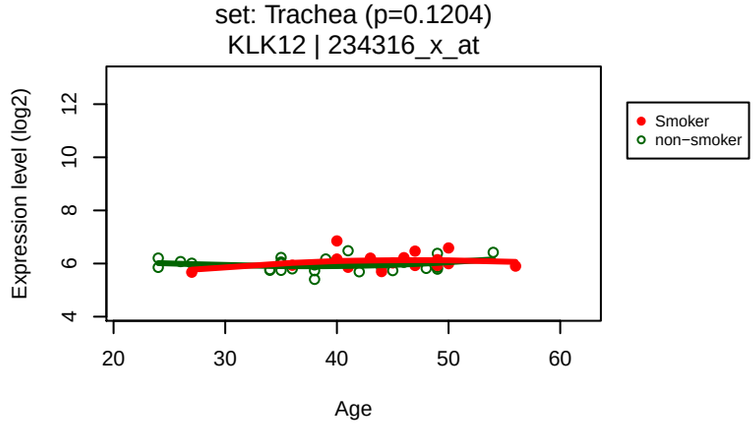

Gene expression ratio in Males
gene: KLK12 | 234316_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1506
p=0.015
p=0.1715
p=0.4643
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KLK12 | 234316_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4172
p=0.5956
p=0.3225
p=0.5792
SmallAirway2
LargeAirway2
Trachea
Macrophage2

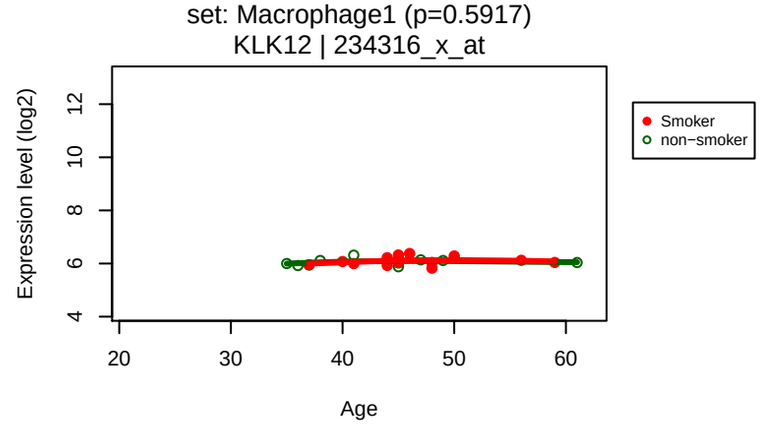

set: SmallAirway1 (p=0.0024)
KLK12 | 234316_x_at
Expression level (log2)
Age
Smoker
non−smoker

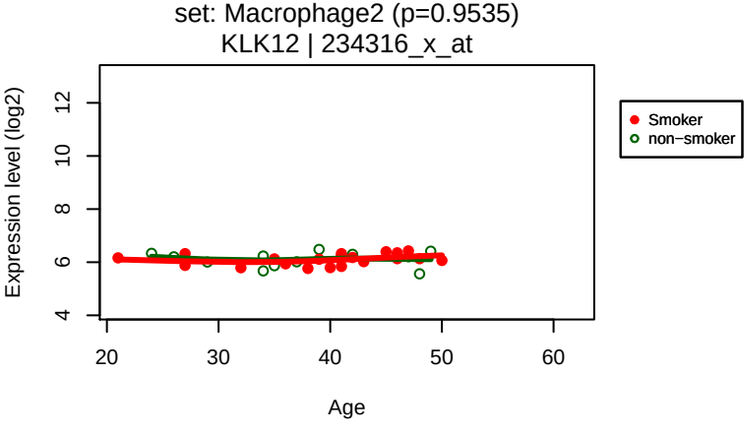

set: SmallAirway2 (p=0.2192)
KLK12 | 234316_x_at
Expression level (log2)
Age
Smoker
non−smoker

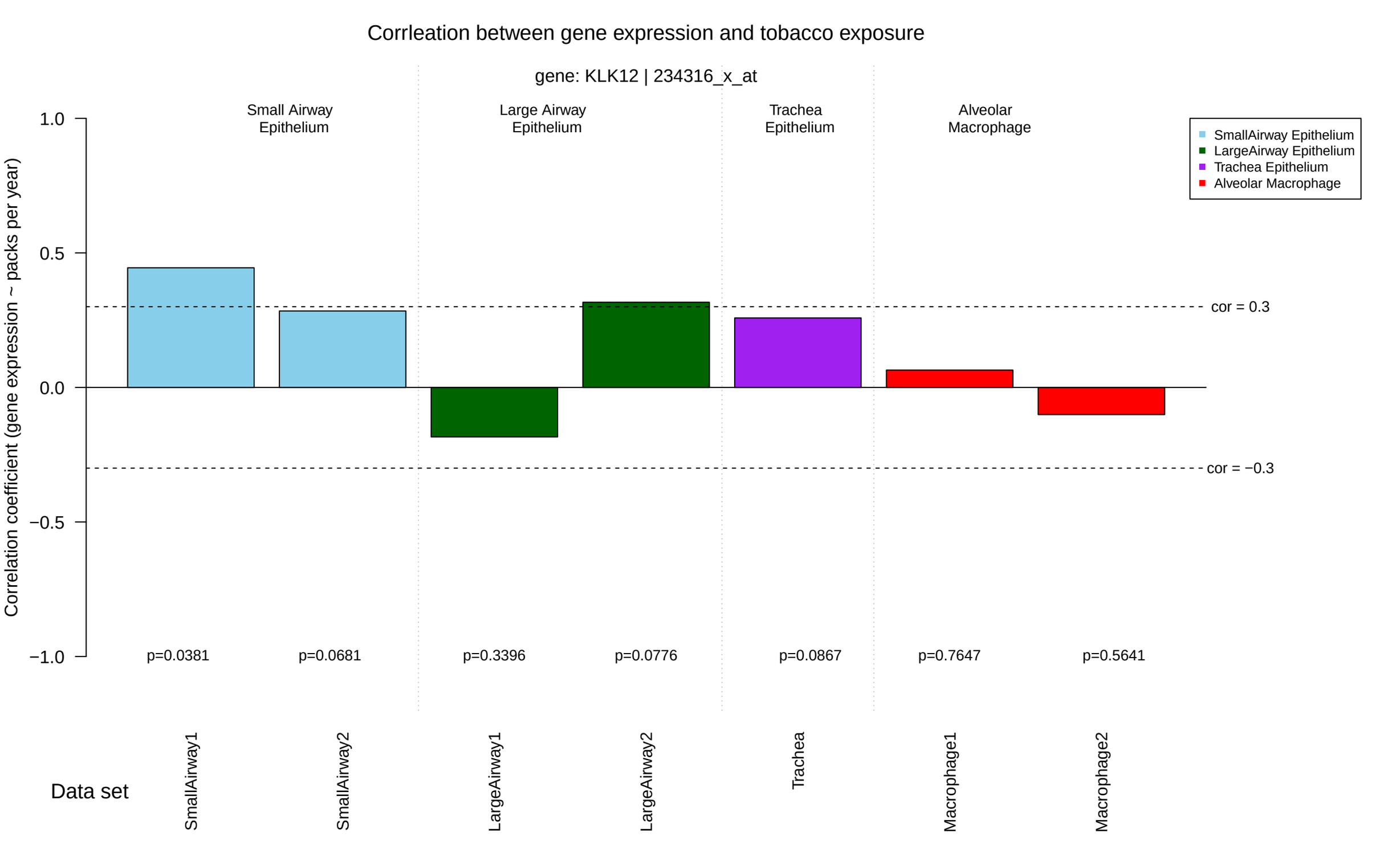

set: LargeAirway1 (p=0.9389)
KLK12 | 234316_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.072)
KLK12 | 234316_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1204)
KLK12 | 234316_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5917)
KLK12 | 234316_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9535)
KLK12 | 234316_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KLK12 | 234316_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0381
p=0.0681
p=0.3396
p=0.0776
p=0.0867
p=0.7647
p=0.5641
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2